

Simulating Exposure for xVA

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Exposure at Default: amount owed by defaulting party at time of default

$$\text{Positive Exposure} = \max(V, 0)$$

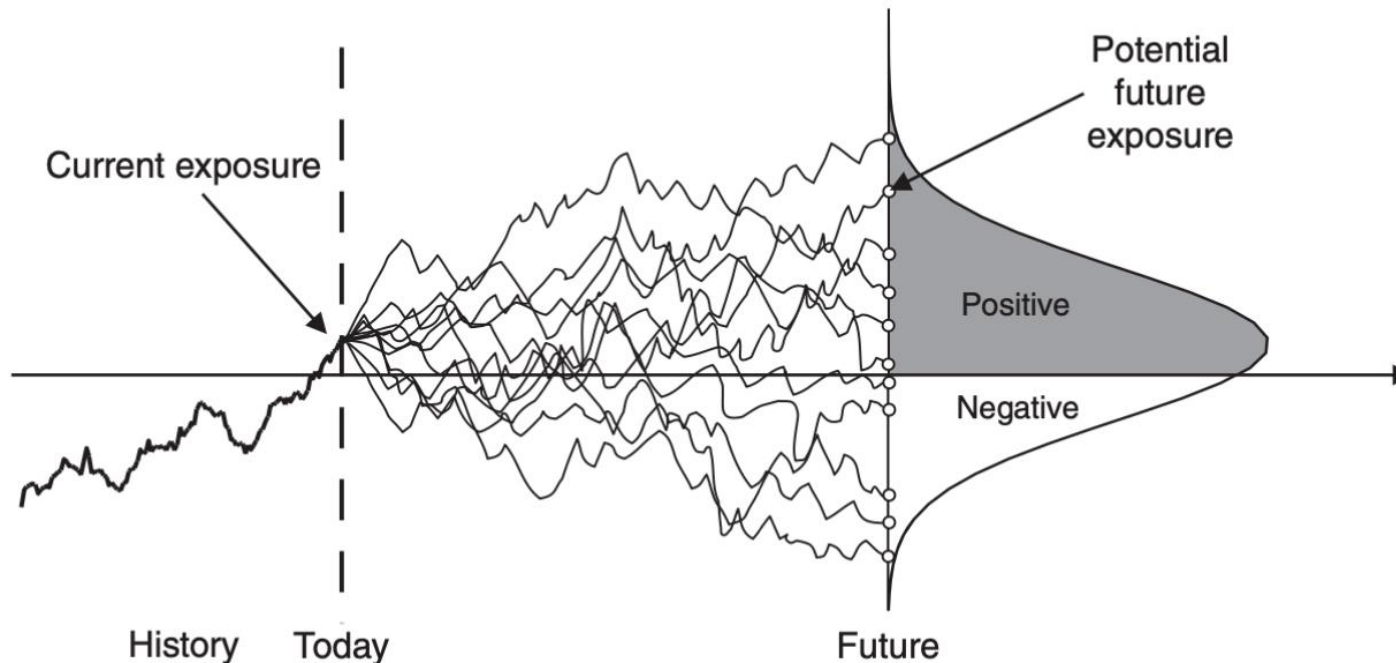
$$\text{Negative Exposure} = \min(V, 0)$$

- Where V is the value of the derivative at the time of default
- Clear asymmetry present

Calculating Exposure

Interest is in future exposure:

- Harder to calculate than current exposure



EPE: Expected Positive Exposure

$$EPE(t) = E [\max(V, 0) \mid \mathcal{F}_t]$$

Keep in mind that derivative V is also an expected value:

$$V(t) = E_t[g(X)]$$

- Which may or may not have a closed form solution
- The nature of the problem is best suited for Monte Carlo simulation

Why Exposure?

Exposure is important in Counterparty Credit Risk:

- For example, consider the unilateral CVA calculation

$$UCVA(t) \approx -LGD \sum_{i=1}^m EPE(t, t_i) \times PD(t_{i-1}, t_i)$$

- EPE at time t is a key parameter of this calculation
- Hence, important for the pricing of bilateral derivatives

General xVA:

$$xVA = \int_0^{\infty} C(t)D(t)E[X(t)]dt$$

Where $X(t)$ is the utilization component

Term	Area	Utilisation component	Calculation
CVA	Credit	EPE	$\max(\text{value}, 0)$
DVA	Credit	ENE	$\min(\text{value}, 0)$
FVA	Funding (and margin)	EFV	value
FCA	Funding (and margin)	EPE	$\max(\text{value}, 0)$
FBA	Funding (and margin)	ENE	$\min(\text{value}, 0)$
KVA	Capital	ECP	Function of value
MVA	Initial margin	EIM	

We want to estimate θ :

$$\begin{aligned}\theta &= \int_{-\infty}^{\infty} f(x)g(x)dx \\ &= \mathbb{E}_X[g(X)]\end{aligned}$$

The Monte Carlo estimate is:

$$\hat{\theta} = \frac{1}{N} \sum_{n=1}^n g(x_n)$$

Where $x_1, x_2, \dots, x_N$ are iid observations of X

Based on the previous idea:

$$E\hat{P}E(t) = \frac{1}{N} \sum_{n=1}^N \max(\hat{V}^{(n)}(t, T), 0)$$

- Time steps: $t \in \{1, 2, \dots, T\}$
- Simulation Paths: $n \in \{1, 2, \dots, N\}$

Step 1: Simulate the states of the world

- Result is an N by T matrix for each risk factor

Step 2: Price derivative V at each state

- Results in an N by T matrix of prices
- For netting set level EPE:

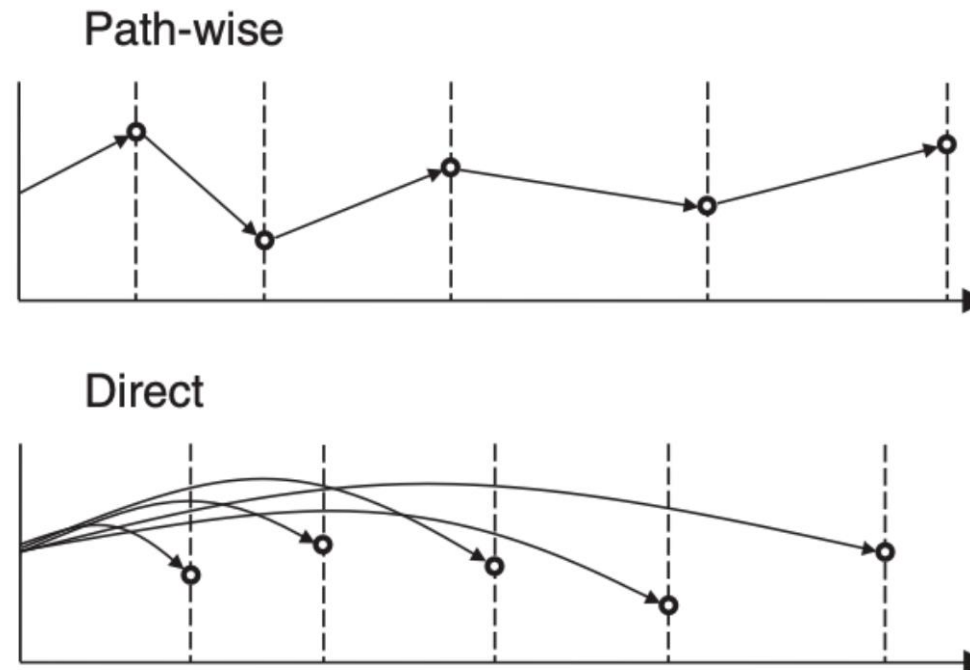
$$V_{\text{Agg}}^{(t)}(t) = \sum_{k=1}^K V^{(n)}(k, t)$$

Step 3: Calculate $\hat{EPE}(t) \forall t \in \{1, 2, \dots, T\}$

- Set all negative values to 0
- Take average over each column
- Results in vector of T EPE values

Complexities of Simulating EPE

- Dimensionality: time complexity of $\mathcal{O}(TNKC)$
- Model Complexity: accuracy v.s. speed
- Pathwise v.s. Direct



Nested Monte Carlo:

- The original MC estimate from before is

$$\theta = E_X[g(X)]$$

- The simulation becomes a nested problem if function g is itself an expected value that requires Monte Carlo to estimate
- In the EPE simulation, this happens when the derivative does not have a closed form solution - examples are American options, Asian options, Bermudan options

Nested Monte Carlo Procedure:

Step 1: Simulate the states of the world

- Result is an N by T matrix for each risk factor - Same as before

Step 2: Price derivative V at each state

- Each state requires its own simulation with N_{inner} paths of $(T-t)$ time steps
- Each inner simulation has a time complexity of $\mathcal{O}((T-t)N_{inner})$

Step 3: Calculate EPE

- Same as before

MC with antithetic variables:

$$\hat{\theta}_a = \frac{1}{n} \sum_{i=1}^n \frac{g(X_i) + g(-X_i)}{2}, \quad \text{with i.i.d. } X_i \sim N(0, 1)$$

- Popular variance reduction technique
- Variance reduction guaranteed if g is monotone increasing

SRM: Method of simulation specifically designed to speed up nested simulations

- Feng and Li (2022)

Step 1: Simulate N_{outer} states of the world - same as before

Step 2: For each time point t

- Select B outer paths from 0 to t - these are the reference paths
- Run the inner simulations with N_{inner} paths for each reference path
 - Save each simulated value at $t+1$; Save needed future path info

Step 2: Continue

- These inner simulations are recycled - no more inner simulations
 - For each new outer path that needs an inner simulation, we assign its nearest reference
 - Rebuild the payoff using the reference paths inner simulations, but with the new paths own past values
 - For each inner path, we also assign a weight
 - Use the weights to get a weighted average derivative value based on each recycled inner path

Step 3: Once repeated for each time step, calculate EPE

Key idea: reuse the inner simulations but weight the results

Weights: Radon-Nikodym Derivative

- The weight used for GBM simulation:

$$w_{i|r,j} = \frac{\exp \left[-\frac{(\ln(Y_j/S_i(t)) - \mu)^2}{2\sigma^2 \Delta t} \right]}{\exp \left[-\frac{(\ln(Y_j/S_r(t)) - \mu)^2}{2\sigma^2 \Delta t} \right]} = \exp \left[-\frac{(\ln(Y_j/S_i(t)) - \mu)^2 - (\ln(Y_j/S_r(t)) - \mu)^2}{2\sigma^2 \Delta t} \right]$$

- Where i is the outer simulation that we are on, r is the reference outer simulation and j is the inner simulation.
- Y_j is the first step for the inner path, $S_i(t)$ is the current price for outer simulation i.

Ways of speeding up simulations:

- Find optimal number of paths for inner and outer simulations
- Vectorization
- Multiprocessing
- Fast programming language e.g. C++

Simulating Risk Factors

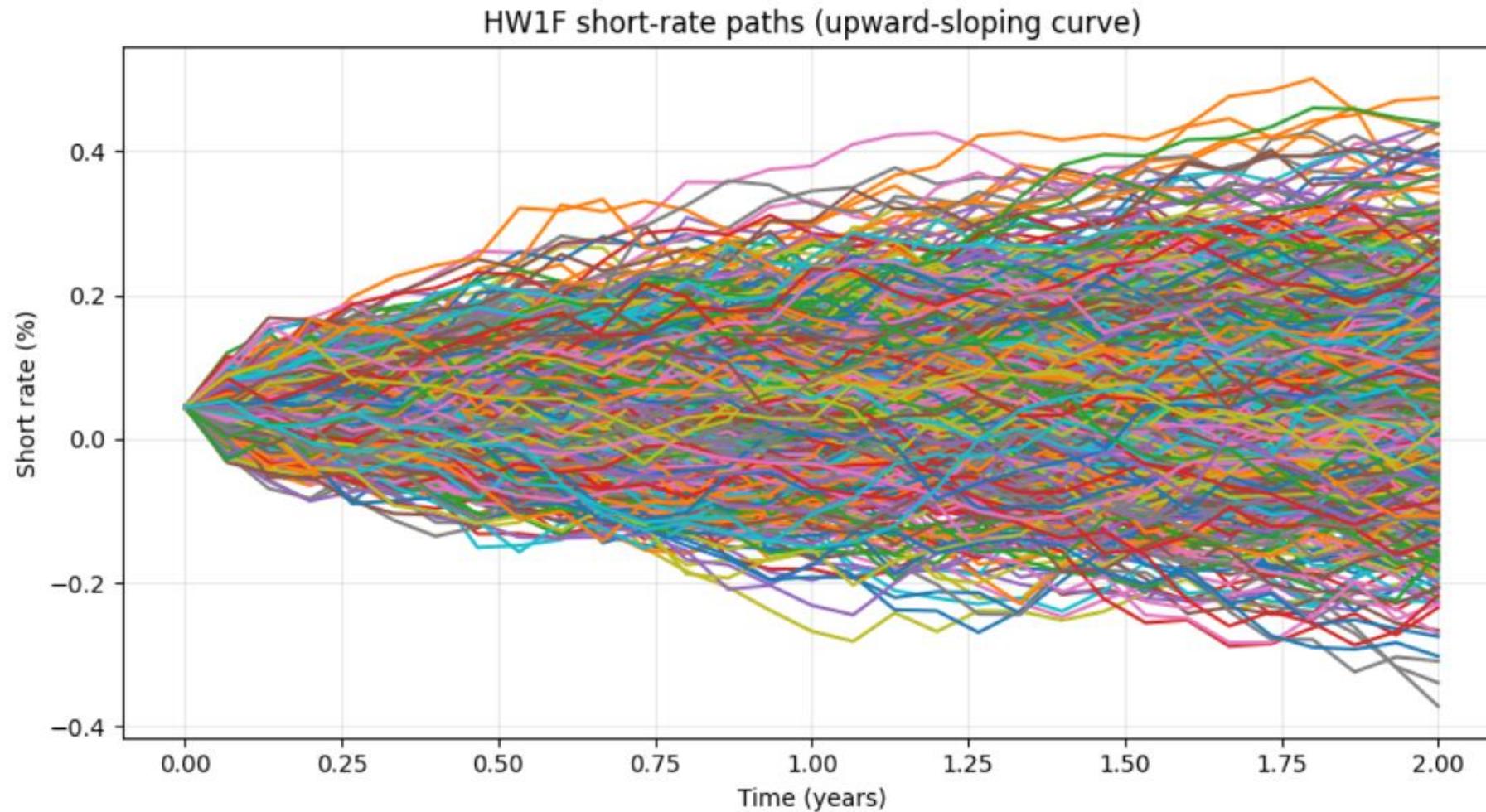
Interest Rates: Hull-White one factor model

$$dr_t = [\theta(t) - ar_t]dt + \sigma(t)dW,$$

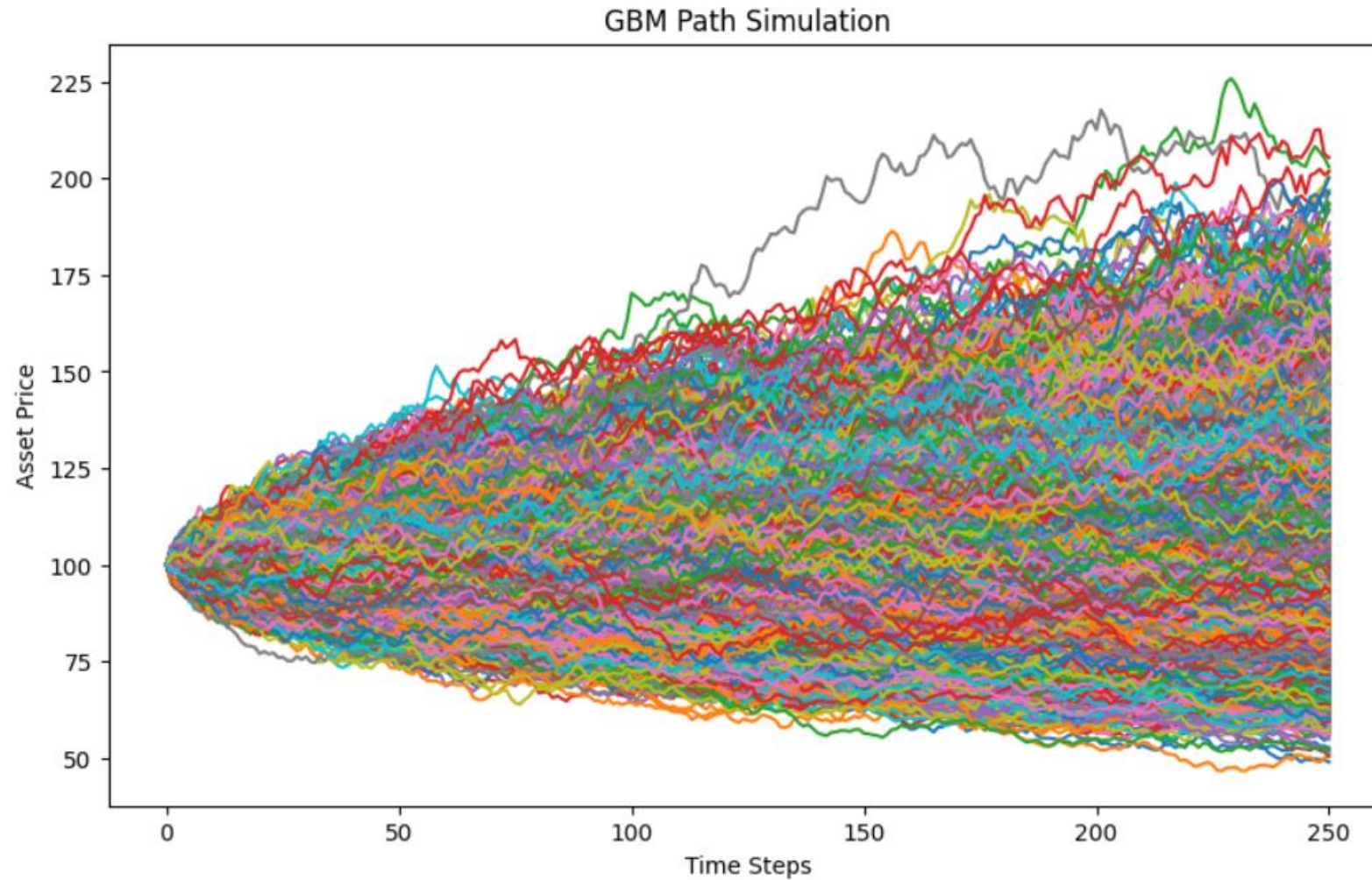
Equity Prices: Geometric Brownian motion

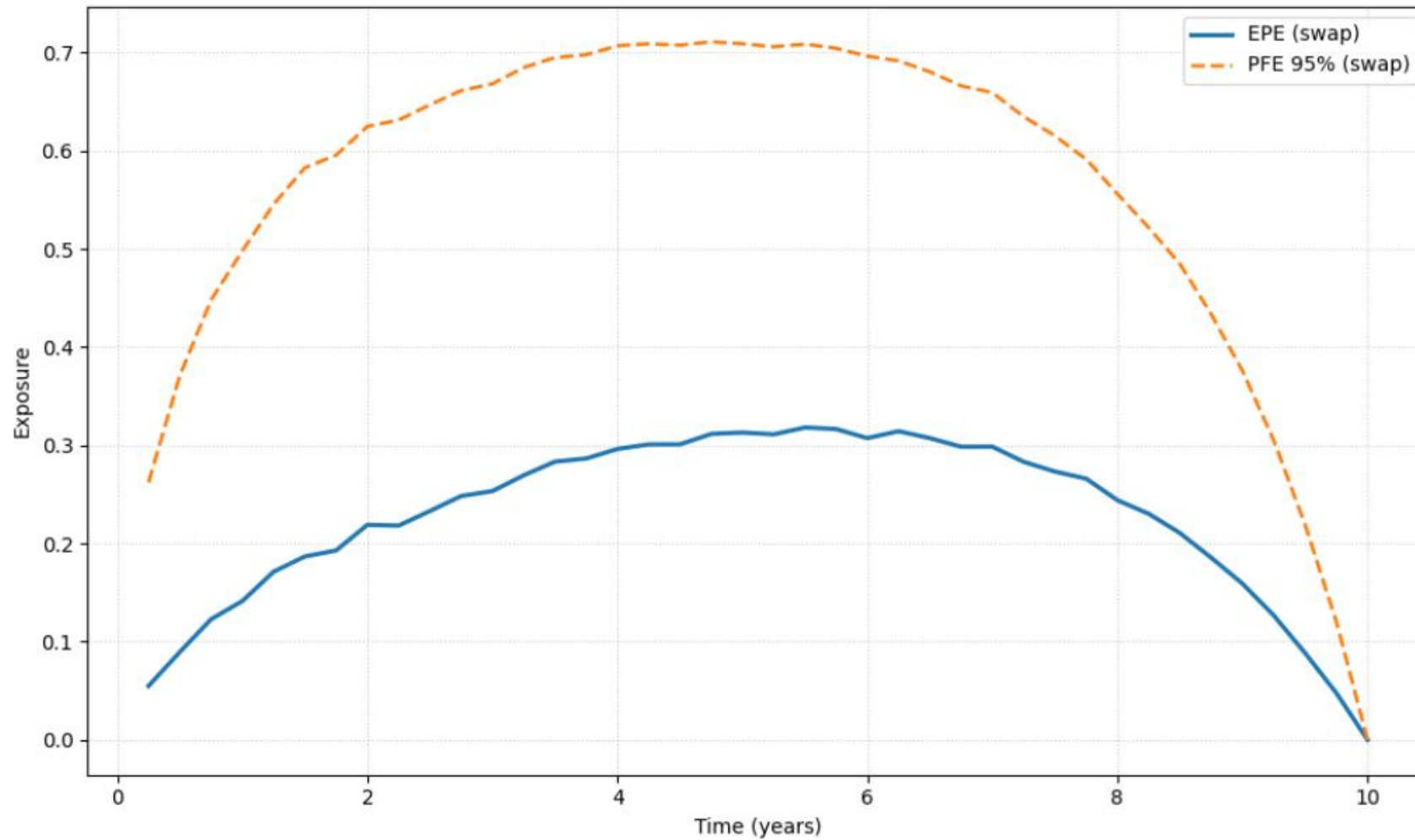
$$dS = rSdt + \sigma SdW$$

Short Rate Simulation



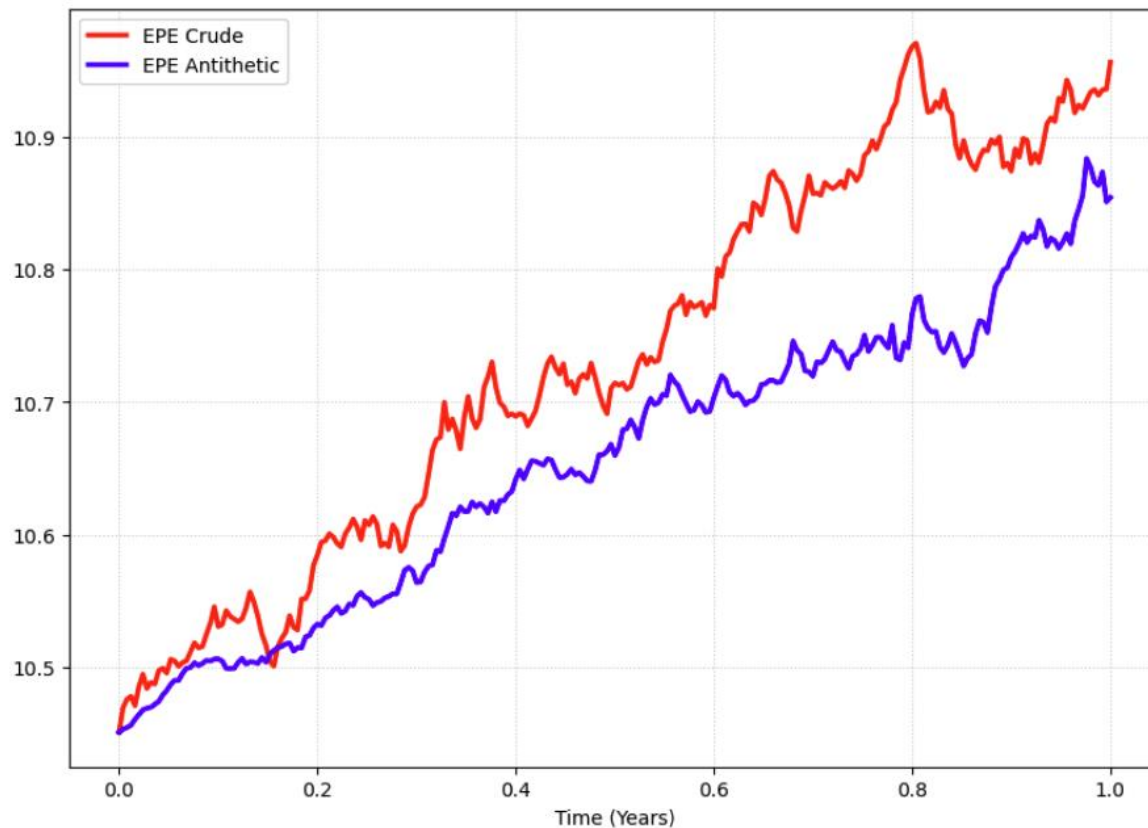
Equity Simulation



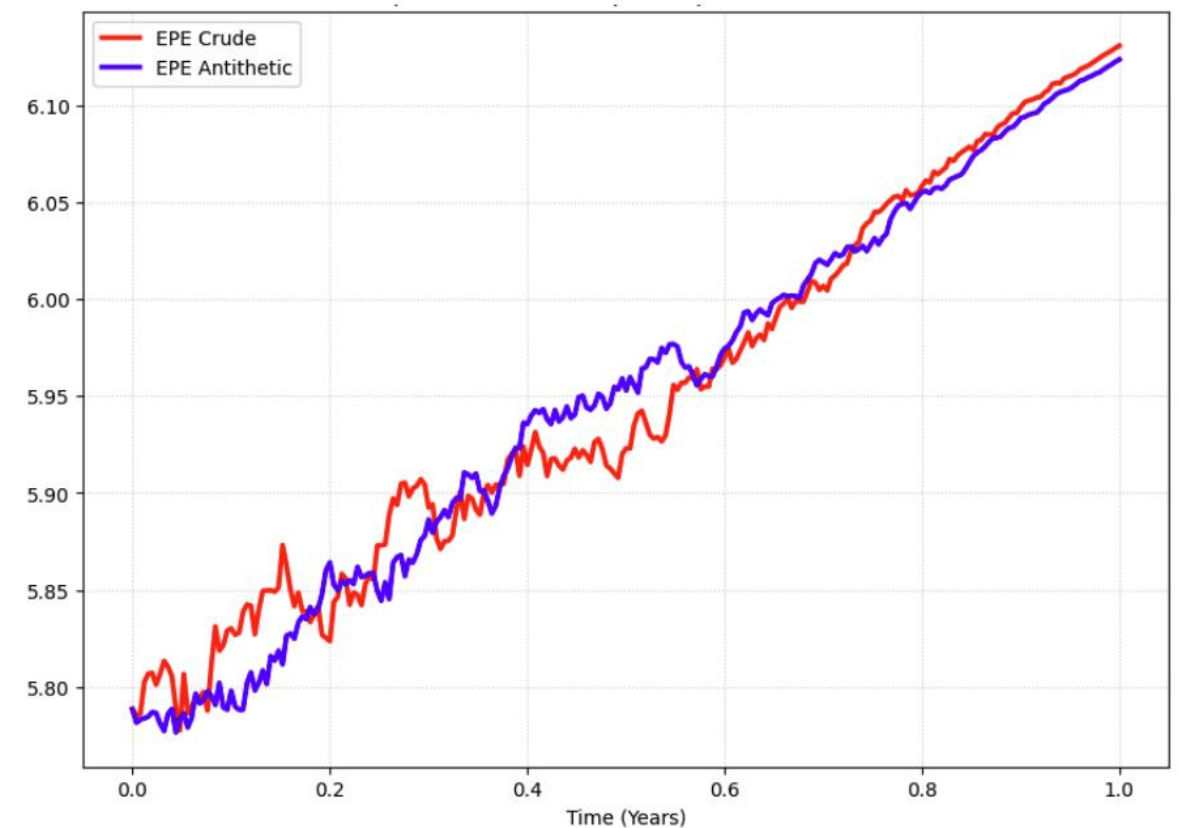


Antithetic Variables

European Option:

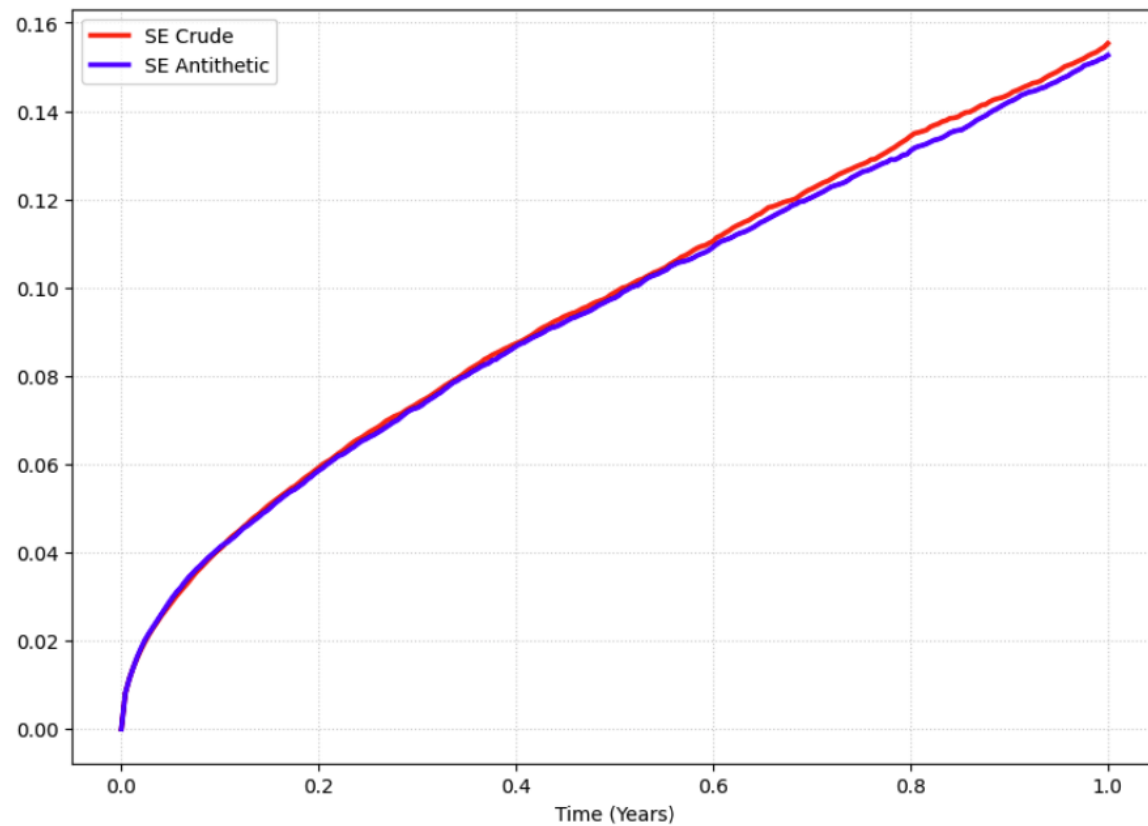


Asian Option:

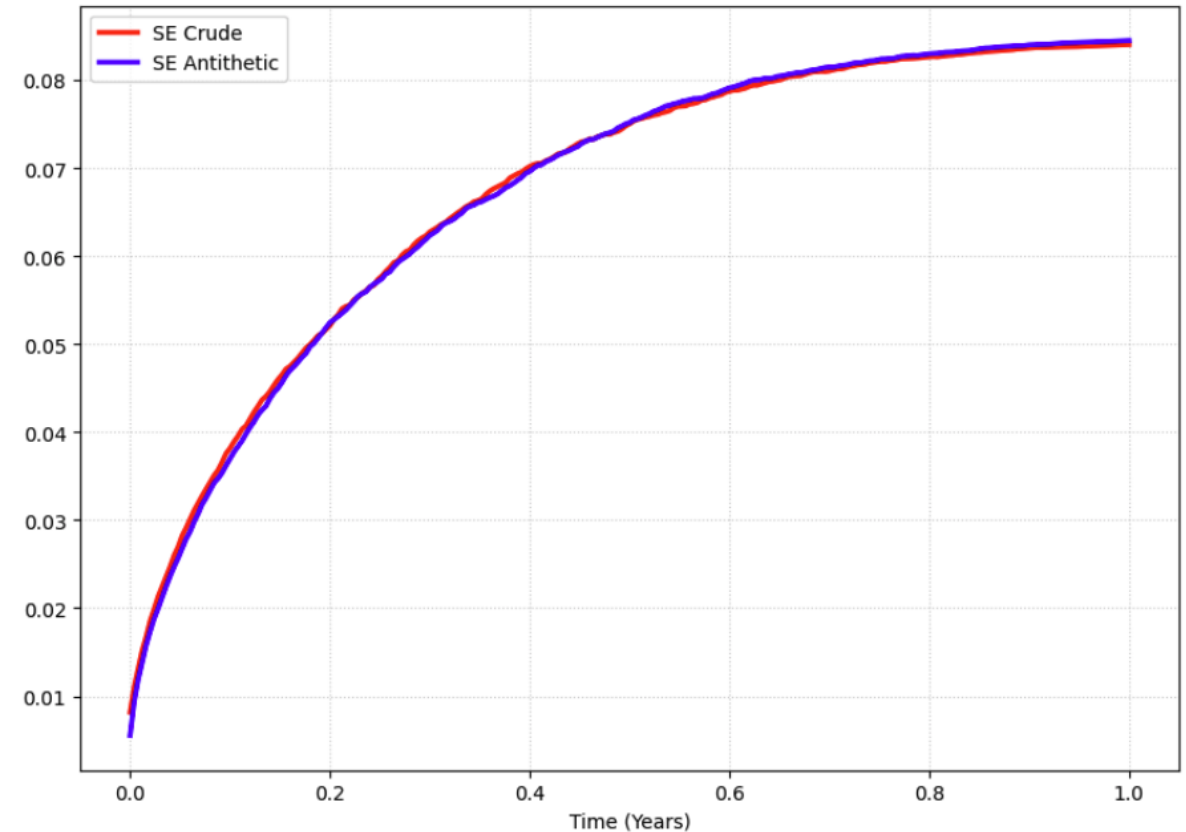


Antithetic Variables

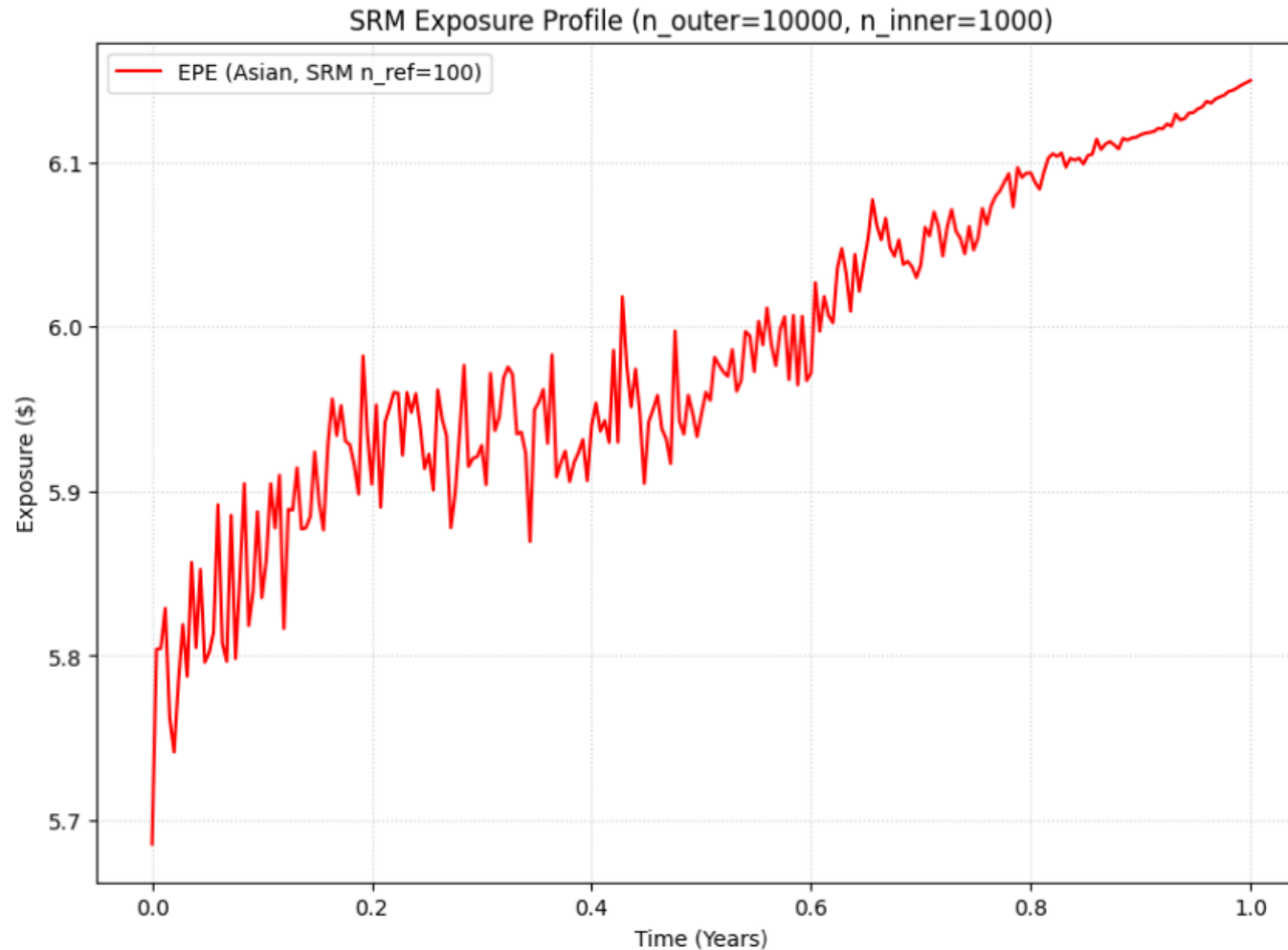
European Option SE:



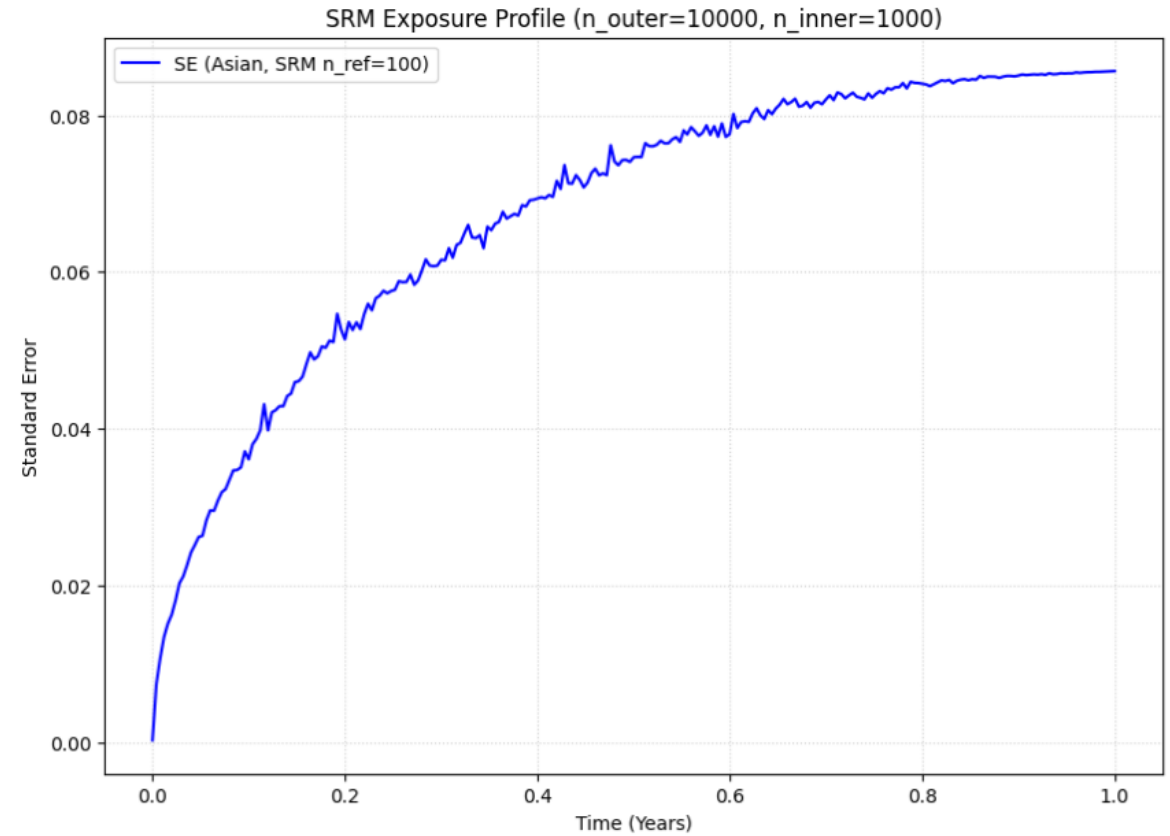
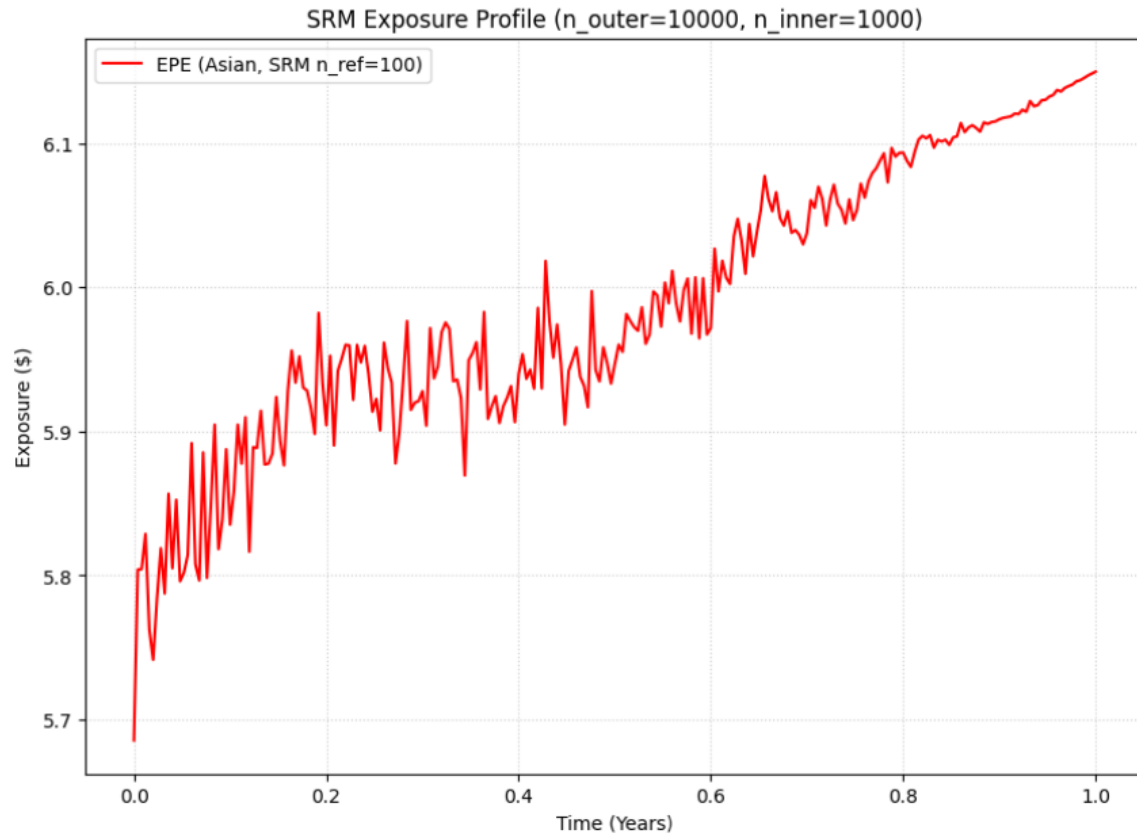
Asian Option SE:



Sample Recycling



Sample Recycling



Vectorization and Multiprocessing

Vectorization:

- EPE simulation for European option
 - 10 000 simulations
 - 250 time steps
- Crude implementation: 125.9185 seconds (just over 2 minutes)
- Vectorized implementation: 0.2146 seconds

Multiprocessing:

- Crude nested for Asian option: 814.0936 seconds
- Parallelised version: 178.62 seconds

Key Points

- The project explored simulating future exposure and its main challenges
- Monte Carlo is accurate but computationally expensive
- Efficiency improves with vectorisation and multiprocessing
- Sample recycling greatly reduces runtime with similar accuracy
- Nested simulations become more practical for complex trades
- Smart algorithms and parallel computing speed up xVA estimation

Thank you
Enkosi
Dankie

